

Personal Statement

In the summer of 2024, as a **data analyst intern at Alibaba**, I wrote retention reports that could say what had happened and could not say why; the questions I was asked were about why, and about what would change under a different decision, and nothing I had could answer them. A year later, as a **machine learning research intern at iFLYTEK**, I trained Transformer and Conformer models on in-house text and speech data that were good on every benchmark we had, and nobody on the team, myself included, could say when they would stop being good. I have come to think of those as the same failure seen from two sides, one about causes and one about distribution shift, and they are the reason this application exists.

The habit those questions demand, staying with a problem until the answer can be defended, is one I learned before I had the questions. My degree was split between Xi'an Jiaotong-Liverpool University and the University of Liverpool, and I arrived in Liverpool behind in academic English; the pace of lectures and the standard of written argument rose at once, and I answered with more hours on both the mathematics and the writing. I graduated with **First Class Honours**, ranked in the **top 1%** of my major.

Sensitivity Analysis for Matched Observational Studies: The causal gap came first. I began a master's in Computational and Applied Mathematics at the **University of Chicago** in September 2025 and, in my first quarter, joined **Prof. Xinran Li** in the Department of Statistics, whose work on sensitivity analysis for matched observational studies is the formal version of Alibaba's question: what would have happened otherwise, and how much hidden bias would it take to overturn the answer. For matched pairs, with Pengyun Wang, I worked out how to centre the within-pair assignment probability at the fitted propensity odds instead of at one half, allow bounded hidden bias around that centre, and build a test of Neyman's weak null that stays valid for every assignment distribution the bound permits. We extended the construction to full matching and applied it to the right heart catheterisation study, where the propensity imbalance left inside the pairs changes how robust the mortality conclusion is reported to be. The manuscript is a **co-first-author working paper**, and it taught me to own a claim: every statement traces back to a distribution over assignments I can defend, under review from a co-author and an advisor who read every line.

Scientific World Models for Research Agents: The distribution-shift gap is the centre of my work with **Prof. Zhuang Liu's** group, which I joined in March 2026 and where I spent the summer of 2026 as a **research intern at Princeton University**. We are building a scientific world model: a predictor that takes the state of a research project and a candidate action and returns the outcome before the experiment is run. Our first predictors did well when replaying past projects but failed on kinds of actions they had never seen, and I designed the repair: predictors that condition on which part of the training pipeline an action changes, and a router that sends each query to the predictor whose experience covers it, or to none. That last option raised a prior question: when should the predictor answer at all? Re-running the 108 doubtful action pairs under several random seeds showed that some orderings were seed noise, and a policy that ranks a pair only when its ordering is stable made **no errors on the 51 pairs it answered** and declined the rest. As a mathematician in a machine learning lab, I learned to say "unsupported" instead of producing a number; I am now building a checker that holds the agent's written reports to the same standard against its own logs.

Two Further Projects: Two further projects sit beside these. With **Prof. Xiao Luo** at the **University of Wisconsin–Madison** I developed an audio-visual test-time adaptation method, the distribution-shift problem in another modality. With **Prof. James Evans** at the **Knowledge Lab** I co-designed the agent-based simulations for a study of how AI breakthroughs narrow scientific

attention, now submitted to *Science*, where the causal question could not be identified from data and the simulation had to isolate the mechanism instead. Across all four, I kept returning to the two questions the internships left me with: how to defend a causal claim, and how to know when a model's answer should be trusted.

Why a PhD at NUS: I am applying to the Department of Statistics and Data Science at NUS because its faculty work on both questions: on the causal side, randomization tests and sensitivity analysis for unmeasured confounding; on the trust side, evaluation under covariate shift and of model-generated output, and conditional inference with auxiliary and synthetic data. They pose the theoretical problems, and their applied studies need the answers; I do my best work when I have both, and a PhD here is the preparation for the faculty position in statistics that I want.